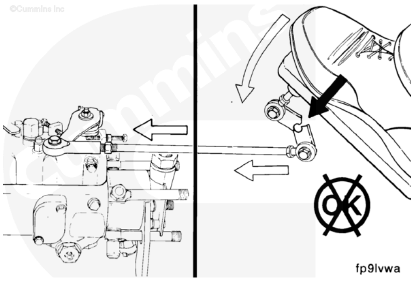
Trainings

Adjust

Fuel Control Lever Travel and Adjustment

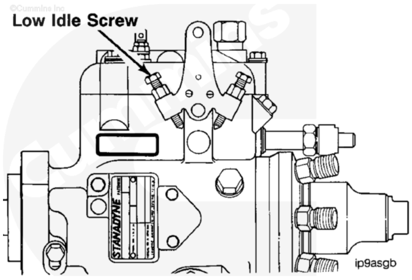
The amount of fuel injected, and subsequently the speed and power from the engine, is controlled by the fuel control lever. Restricted travel of the lever can cause low power. **Always** check for full travel of the lever when diagnosing a low-power complaint.



Stanadyne DB4 Fuel Injection Pump Adjustment Screw

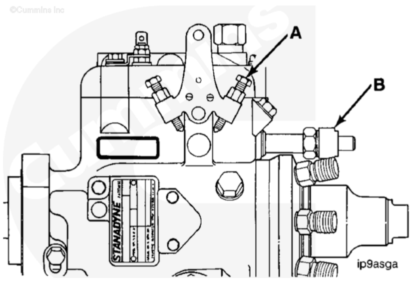
The low-idle adjustment screw on the DB4 fuel injection pump is mounted on the control lever assembly. The adjustment screw can be used to increase the idle speed to compensate for accessory loading. The low-idle adjustment screw **must** be adjusted by an authorized service dealer and resealed.

Note : Never turn the idle adjusting screw out (reduce idle speed) on the speed drop governor-equipped fuel injection pump; this can result in disengagement of the throttle lever from the guide bushing.



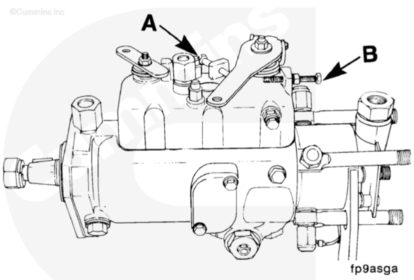
The high-idle adjustment screw (A) is sealed. The adjustment screw on the DB4 fuel injection pump is mounted on the control lever assembly. The high-idle adjustment screw **must** be adjusted by an authorized service dealer.

The speed droop adjustment screw (B) is located above the delivery head. The fuel pump governor sensitivity can be adjusted to increase or decrease governor regulation.



Lucas CAV DPA/DPS and Delphi DP210 Fuel Injection Pump Adjustment Screws

The idle adjustment screw provides a stop for the lever at low speed. The adjustment screw can be used to increase idle speed for accessory loading, or, if required, to lower the idle



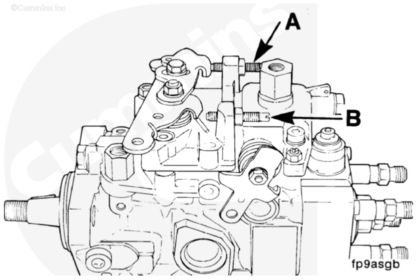
speed. The high-idle screw is sealed and **must** be adjusted by an authorized repair shop, and then resealed.

A - Idle screw

B - High-idle screw

⚠ CAUTION ⚠

The fuel control lever on the Bosch® VE fuel injection pump is indexed to the shaft during pump calibration. If the lever has been removed and reinstalled incorrectly, engine speed and power will be affected.



Bosch® VE Fuel Injection Pump Adjustment Screws

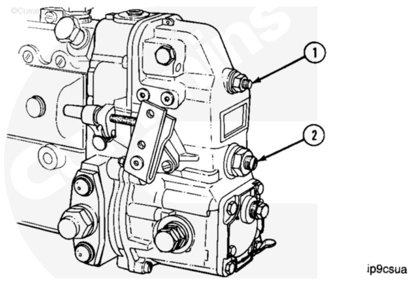
A - Idle Screw

B - High-Idle Screw

The high-speed adjustment screw on both fuel injection pumps provides the stop for full speed. The high-speed adjusting screws are sealed. Adjustment of this screw **must** be performed **only** by an authorized fuel injection pump service center, and then resealed.

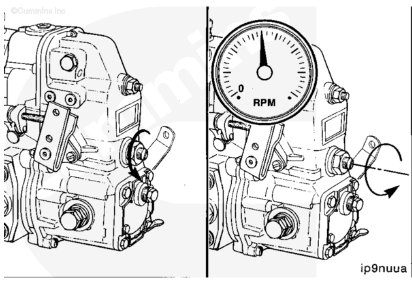
The high-speed adjusting screw can be used to derate engines.

Bosch® RSV Governor

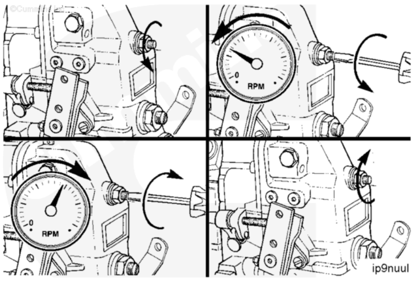


Idle speed adjustment for industrial engines requires the setting of both the low-idle speed screw (1) and the bumper spring screw (2).

First, loosen the locknut; then, back out the bumper spring screw until there is no change in engine speed.



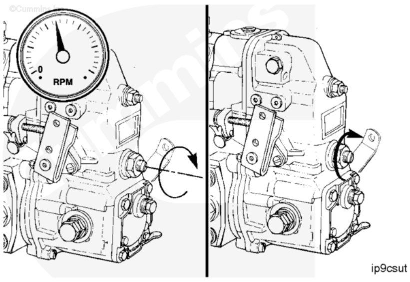
Loosen the locknut, and adjust the idle speed screw to 40 to 50 rpm less than the desired speed. Turn the idle speed screw **counterclockwise** to decrease rpm and **clockwise** to increase rpm.



Tighten the locknut.

Torque Value: 8 n•m [71 in-lb]

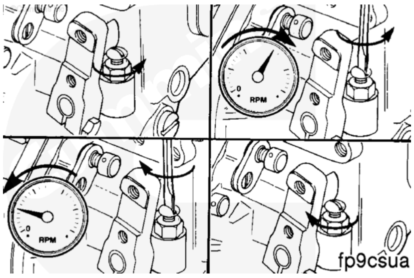
Turn the bumper spring screw **clockwise** until the desired idle speed is obtained.



Tighten the locknut.

Torque Value: 8 n•m [71 in-lb]

Bosch® RQV and RQV-K Governor



Idle speed adjustment on automotive fuel injection pumps requires setting of the stop screw.

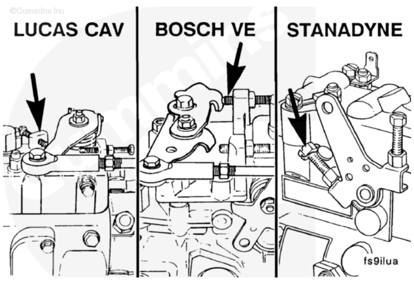
Loosen the locknut, and turn the idle speed screw **counterclockwise** to increase the rpm and **clockwise** to decrease the rpm speed.

Tighten the locknut.

Torque Value: 8 n•m [71 in-lb]

⚠ CAUTION ⚠

Do not reduce idle speed from factory setting on the Stanadyne DB4 fuel injection pump. Internal damage can result.



Distributor Pumps

Loosen the adjusting screw locknut, and adjust the idle as required.